

INSTITUTE OF ACTUARIES OF INDIA

EXAMINATIONS

16th March 2018

Subject SA6 – Investment

Time allowed: Three Hours (14.45* – 18.00 Hours)

Total Marks: 100

INSTRUCTIONS TO THE CANDIDATES

1. *Please read the instructions inside the cover page of answer booklet and instructions to examinees sent along with hall ticket carefully and follow without exception.*
2. ** You have 15 minutes at the start of the examination in which you are required to read the questions. You are strongly encouraged to use this time for reading only, but notes may be made. You then have three hours to complete the paper.*
3. *You must not start writing your answers in the answer sheet until instructed to do so by the supervisor.*
4. *The answers are expected to be India Specific application for the syllabus and corresponding core reading. However, substantially the core reading material is still taken from material supplied by Actuarial Education Company which is meant for UK Fellowship examination. The core reading also contains some material which is India Specific, mostly the IRDA regulation. In view of this, it should be noted that focal point of answers is expected to be India Specific application. However, if application specific to any other country is quoted in the answer the candidate should answer the question with reference to Indian environment.*
5. *Attempt all questions, beginning your answer to each question on a separate sheet.*
6. *Mark allocations are shown in brackets.*
7. *Please check if you have received complete Question paper and no page is missing. If so, kindly get a new set of Question paper from the Invigilator.*

AT THE END OF THE EXAMINATION

Please return your answer booklet and this question paper to the supervisor separately. You are not allowed to carry the question paper in any form with you.

Q. 1) The banking sector Non Performing Assets have been growing. Recently the Government announced recapitalization of the banks of INR 2.11 lakh crore to be infused over the next 2 years into public sector banks.

i) Define bank recapitalization? (3)

Further it has been announced that 1.35 lakh crore of this recapitalization will be in the form of recapitalization bonds and remaining 0.76 lakh crore will come from budgetary allocation and fund-raising from the markets.

ii) What are the alternate methods available to the Government to recapitalize? What is the main challenge with alternate method? (4)

iii) What are recapitalization bonds and how will the recapitalization bond issue work? (4)

iv) What would be the impact on the fiscal deficit, exchange rate and overall impact on the economy? (12)

v) What is the impact of this method of recapitalization? (2)
[25]

Q. 2) You are an Investment consultant to a small defined benefit Pension scheme.

The trustees have asked you to give advice on investment strategy of the Pension fund.

i) What are the considerations that need to be taken into account when formulating an investment strategy for a pension scheme? (5)

Recently the central bank has lowered the interest rates causing the yields on bond portfolio to drop. You have been asked to look for other investment options to boost the fund performance.

The trustees have in particular asked you to look at exploring the investment opportunities in hedge funds. The trustees are aware that historically hedge funds have produced high absolute returns; however there have been cases in which losses also occur. Due to very high volatility in returns, the trustees have sought your expertise.

ii) Why should trustees consider investment in hedge funds? State reasons and explain. (5)

iii) Describe the risks and obstacles of creating a sensible diversified portfolio of hedge funds (5)

In view of the complexity of strategies employed, and lack of transparency and accessibility, evaluating and selecting a hedge fund manager can be extremely difficult. There are two ways to gain access to hedge funds: investing directly in individual hedge funds and allocating capital to a fund of hedge funds.

iv) Please state the hedge-fund specific factors the trustees need to take into account to select a hedge fund manager. (5)

The other concern faced by trustees has been deficit in the fund which has been caused due to onerous Regulatory changes on solvency capital. This has made it difficult for the fund to repair the deficit. The trustees are looking at increasing investment in equity, however since the size of pension fund is small you feel that volatility of equity investment could be risky for the fund.

You feel that the Pension fund can look at investment in equity derivative to achieve the required goal or alternatively need to look at the derivatives to hedge the equity volatility.

- v) Describe why a derivative based product may be a suitable investment for the pension scheme trustees to consider in helping to reduce the deficit (8)
[28]

Q. 3) You are the Consulting Actuary to a large retail bank that has a huge book of mortgages and credit card receivables. The Bank, in its recently concluded Asset Liability Committee meeting, has decided to take the help of your firm to advise them on Asset backed securities and what options are available for them.

- i) Explain what is meant by Asset backed securities/securitizations. (2)
- ii) Outline the key risks (3)
- iii) Explain generally how an Asset Backed security can be structured and the key risks involved (5)
- iv) Explain in detail how a Collateralised Debt Obligation (CDO) can be structured. (5)
[15]

Q. 4) Recently Product Regulations review committee submitted its recommendation on various changes required. Included in them was a section on reform of investment norms. Given below are the main recommendations.

- Lower the mandatory proportion of 'G-Secs' in the Life Fund and the Pension & General Annuity Funds and allow for higher exposure in alternative higher yielding assets (e.g. equity or property) or high rated corporate bonds
- Currently, insurers are allowed to use interest rate derivatives for hedging future cash flows from contracted policies. This can be widened to allow usage of other derivatives such as options on bonds and equity and futures on equity.

- i) You are the CIO and your CEO has asked you to draft a note on what factors to consider which will impact your business. In your response also add in your viewpoint of current situation of the industry and why in your opinion these recommendations have been made.

In drafting your response you can assume that you are from a mid-size company which has been launched about 10 years ago. Portfolio is 65% Par, 30% Non-Par and 5% Unit Link with Guarantee. Your company has very limited surplus and it has been struggling to sell par products in the last 2 years. Often the returns get compared to the performance of the mutual fund industry.

(10)

- ii) Given this possibility of being able to invest in interest rate derivatives your product actuary would like to introduce a guaranteed equity product.

This product will promise to pay return equal to 95% of the increase in capital value of Nifty 50 index over a 5 year period together with a guarantee to return your initial investment should the index fall over 5 year period.

Outline how you may be able to hedge the liability of such a fund?

(2)
[12]

Q. 5) Your young artist friend in his early thirties has recently sold one of his painting for INR 50 lakhs. He has no liabilities. He would like to invest the money in a mutual fund but is very confused with the various types of Mutual funds being offered in the market.

- i)** Compare the following type of Mutual funds –
Growth Funds, Income Fund and Money Market funds by their
- a) Objective
 - b) Investment Pattern
 - c) Suitability for him. (5)
- ii)** Discuss the factors that he should consider while selecting a Mutual funds? (5)

You also consult with another AIF. Your consultancy's senior partner has asked you to comment on which AIF is performing better for recommendation to another client who has approached him. You have the below information with you over a 5 year period. The annual risk free rate of return over this period is 5% per annum.

	Fund X	Fund Y	Index
	% p.a.	%p.a.	%p.a.
Annual Return	20%	14%	8%
Standard Deviation	25%	20%	12%
Correlation Co-efficient	0.8	0.4	1
Tracking Error	6%	2%	

- iii)** Calculate and comment different risk adjusted performance measures for each hedge fund. (5)
- iv)** Comment on the results from part (iii), stating any limitations that apply to them. (5)

[20]
